AGRICARBON PRACTICES

PREMISE

This practice was developed to provide a **comprehensive and rigorous framework for quantifying, monitoring and reporting greenhouse gas (GHG) emissions and removals in agricultural contexts**Its creation responds to the growing demand for reliable and transparent tools for assessing the environmental impact of agricultural practices. The main objective is to **support the transition towards more sustainable agricultural systems** through accurate and transparent quantification of GHG emissions and removals, continuous monitoring of environmental performance, and clear and consistent reporting of results. The scope of application focuses on agricultural production systems that include soil management, such as the cultivation of field and tree crops, grassland management, and agroforestry, and that generate net emissions or net removals of GHGs from the atmosphere.

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INTRODUCTION

In this context, the **verification and validation** emerge as key processes for **ensure integrity and credibility** of carbon credits generated by the projects. A central role is played by the**Validation and Verification Body (VVB)**, independent bodies that operate independently and competently in these phases.**validation (ex-ante)**It takes place before project implementation, with the aim of evaluating the reasonableness of the assumptions, limitations, and methods that support statements about the outcome of future activities. Subsequently, the**verify (ex-post)**Evaluates the declarations with historical data and information to determine whether they are factually accurate and compliant with established criteria, then takes action after the project has been implemented. This dual review phase is essential to instill trust and ensure the realization of tangible climate benefits in the voluntary carbon market.

PURPOSE AND SCOPE

This document provides project-level guidance for creating activities intended to generate emission reductions and increases in greenhouse gas (GHG) removals and specifies the actions to be taken to certify these activities and generate carbon credits.
It includes requirements for planning a GHG project, identifying and selecting GHG sources, sinks and reservoirs (SSRs) relevant to the baseline and the project, monitoring, quantifying, documenting and reporting GHG project performance, managing data quality as well as third-party verification of data.

This document has been developed to outline the **fundamental requirements and procedures for the rigorous verification and validation of projects** that conform to the**Metodologia e allo Standard TCR (Trusted Carbon Reduction)**. Tale**methodology, validated by RINA**, provides a comprehensive and scientifically robust framework for the **quantification, monitoring and reporting of greenhouse gas (GHG) emissions and removals in agricultural contexts**.

Purpose of this Guide:

The main purpose of this Guide is to provide a practical and transparent tool for:

- **Determine the project boundaries** in accordance with TCR criteria.
- **Identify and validate the baseline scenario**, which represents conventional farming practices and the GHG emissions/removals that would occur in the absence of the project intervention.
- **Evaluate the additionality of the project**, ensuring that the GHG reductions or removals generated are actually additional to what would happen in the "business-as-usual" (BAU) scenario.
- **Monitor relevant parameters**(such as Soil Organic Carbon - SOC, biomass and emissions) continuously and reliably.
- **Accurately quantify GHG reductions or removals** resulting from afforestation, reforestation and the adoption of sustainable agricultural practices.

Scope of Application:

This Guide is applicable to all **agricultural production systems** who intend to implement and certify GHG mitigation and carbon sequestration practices, adopting the TCR (Trusted Carbon Reduction) methodology. Specifically, it is aimed at projects that:

- **They include soil management**: They are designed for activities such as growing grass and tree crops, managing grasslands and pastures, and agroforestry.
- **Generate net emissions and/or net removals of GHGs**:
    - **Net GHG reductions**They occur when the implementation of the agricultural system produces lower emissions than the baseline (for example, switching from traditional ploughing to minimum tillage).
    - **Net GHG removals**They occur when the agricultural system's capacity to remove greenhouse gases exceeds that of the baseline, absorbing a greater quantity of CO_2 from the atmosphere (for example, through the integration of hedges and trees in cultivated fields).
- They are interested in **quantifying and monitoring your environmental impact** in terms of GHG emissions and removals, and to track progress over time.

REGULATORY AND LEGISLATIVE REFERENCES

The credibility and effectiveness of greenhouse gas (GHG) reduction and removal projects in agriculture depend largely on their **strict adherence to an internationally and nationally recognized regulatory and standardized framework**This "Design Verification and Validation Guide" is firmly rooted in these references to ensure maximum integrity and transparency.

At the international level, the document aligns with the **IPCC Guidelines for National Greenhouse Gas Inventories**, including the 2019 refinements, providing a robust scientific basis for quantifying emissions and removals. A central role is played by the **standard ISO**, such as the**UNI EN ISO 14064-2** for quantification, monitoring and reporting at project level, the**ISO 14065** for the accreditation of verification and validation bodies (VVB), and the**ISO 14067**and**UNI EN ISO 14066** for specific skills in the AFOLU sector (Agriculture, Forestry and Other Land Uses). Furthermore, reference is made to **ISO 18400-104** for soil sampling protocols and **UNI EN ISO/IEC 17025** for laboratory analysis.

The methodology incorporates specific methodologies developed by recognized programs, such as **Verra VM0042 v2.0** for the improvement of agricultural land management and the**"AFOLU Non-Permanence Risk Tool" Version 4.2 is coming** for assessing the risk of non-permanence. For afforestation and reforestation activities, the methodology is adopted **CDM AR-AMS0007 v3.1**of the UNFCCC Clean Development Mechanism (CDM). The interoperability of the TCR is ensured by alignment with standards such as **Verra VCS** and **Gold Standard**.

On the national and European front, the document integrates the principles of **European framework LIFE C-Farms**, and complies with specific regulations such as the **EU Regulation 2018/848** for the definition of organic farming. The following are also considered: **MASAF National Guidelines for Agronomic Techniques (2023)** and the regulations of the **Lombardy Region.** The permanence of climate benefits is strengthened by the need for **legally binding agreements** or **legal requirements** for the continuity of management practices, and the additionality of the project is also assessed in reference to the inclusion of AFOLU commitments in the **Nationally Determined Contributions (NDC)** and in the climate plans of the countries that signed the Paris Agreement. This integrated approach is essential to ensure that each carbon credit generated is a true and verifiable reflection of a real and lasting climate benefit.

This set of regulatory and legislative references acts as a legal compass, guiding projects through the complexities of the carbon market and ensuring that the actions taken are valid, measurable, and compliant with global sustainability expectations.

TERMS AND DEFINITIONS

1. **Additionality**: Principle which establishes that the **reductions or removals of Greenhouse Gases (GHG)** generated by a project or activity are **additional** compared to those that would have occurred in the absence of the intervention ("business-as-usual"). The TCR methodology describes additionality tests, including financial additionality, and demonstrates that the project goes beyond common situations.
2. **Organic Farming**: Agricultural production system that **excludes the use of synthetic inputs** such as fertilizers and pesticides, promoting biodiversity, biological cycles and soil health.
3. **Annual (Annual)**: A plant species that completes its life cycle, reproduces, and dies within a year.
4. **Conservative Ploughing**: Soil working techniques that **minimize soil disturbance**, such as zero-tillage and minimum tillage.
5. **Greenhouse Gas Absorber (GHG Absorber)**: Process that **removes a GHG from the atmosphere**. An element that contributes to GHG removals.
6. **Baseline (Reference Scenario)**: Quantitative reference of the **GHG emissions and/or removals** that would have occurred in the absence of the GHG project and provides the reference scenario for comparison with the projected GHG emissions and/or removals. It represents the emissions scenario expected in the absence of the project intervention. In agriculture, it is the carbon removal performance that would occur in the absence of carbon farming practices.
7. **Biofuels**: Type of fossil fuel.
8. **Biochar**: Inert solid material (stable carbon) obtained from **pyrolysis of biomass in the absence or near-absence of oxygen.** It can be used as a soil conditioner to increase soil fertility and sequester carbon.
9. **Biomass**Organic material derived from plants and animals. Includes aboveground biomass (trunks, branches, leaves) and belowground biomass (roots).
10. **Buffer**: Risk of non-permanence of the credit-generating project, expressed as a percentage of credits. In the TCR, a **Permanence Buffer is** a precautionary reserve of credits or sequestered carbon, intended to cover any future losses due to unforeseen events, such as natural disasters or forest fires, ensuring the environmental robustness and continuity of the project.
11. **Soil Organic Carbon (SOC)**: Main carbon sink affected by carbon-farming activities that is expected to increase under the project scenario.
12. **Carbon Accounting**: Process of quantifying the removals or reductions of CO_2 equivalent (CO_2e) achieved through project interventions, particularly in afforestation and reforestation (A/R) projects, according to standardized methodologies.
13. **Intercropping**: Agricultural practice that involves the cultivation of **two or more crops in the same field** at the same time.
14. **Ecological Corridor**: Area of a habitat that **connects separate biological populations** from human activity, through barriers such as roads or houses. It can be created through the planting and/or improved management of green areas and trees outside the forest.

15. **Carbon Credits**: Unit representing a **metric ton of CO_2 equivalent removed, reduced or avoided** through certified activities. Credits can be traded or withdrawn to offset emissions produced elsewhere.
16. **Crediting Period**: Maximum period during which project activities can generate credits.
17. **Ground Data / Field Measurements**: Any data relating to the physical and biological characteristics of trees, shrubs or bushes within an area, or a parcel of land with known and georeferenced boundaries, acquired through direct measurement in the field by a qualified technician or estimated from such measurements.
18. **Apparent Density**: Soil parameter for estimating organic carbon, measured by the dry weight of soil in a known volume.
19. **Diameter at the Breast of a Man (DBH)**: Measurement of the diameter of a tree trunk at 1.37 m above the ground, used to estimate biomass and carbon.
20. **GHG Statement**: A factual and objective, evidence-based statement regarding emissions, removals, reductions or increases in removals of GHGs that is subject to verification or validation.
21. **Anaerobic digestate**: Semi-liquid by-product of biogas production, rich in nutrients and usable as an organic fertilizer.
22. **Soil Disturbance**Any activity that causes a decrease in soil organic carbon (SOC), such as plowing or digging trenches.
23. **Greenhouse Gas Emissions (GHG Emissions)**: Release of a GHG into the atmosphere.
24. **Greenhouse Gas Emission Factor (GHG Emission Factor)**: Coefficient that quantifies the GHG emissions associated with specific activities or processes.
25. **Greenhouse Gas Removal Factor (GHG Removal Factor)**: Coefficient relating GHG activity data to GHG removal.
26. **Synthetic Fertilizers**Industrially produced fertilizers containing nutrients in chemical form.
27. **Urban Forestry (Urban Forestry Project)**: Set of operations for planting and maintaining new public greenery to **permanently increase the carbon stock**, also considering the GHG emissions resulting from such operations.
28. **Greenhouse Gases (GHG)**: A gaseous constituent of the atmosphere, both natural and anthropogenic, that absorbs and emits radiation at specific wavelengths within the infrared radiation spectrum. They include carbon dioxide (CO2), methane (CH4), nitrous oxide (N2O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), and sulfur hexafluoride (SF6).
29. **Ground Truthing**: Field surveys used to collect and record land use information and to serve as independent data for remote sensing classification.
30. **Uncertainty**: Parameter associated with the result of a quantification that characterizes the dispersion of values reasonably attributable to the quantified value.
31. **Topographic Wetness Index (TWI)**: Hydrogeological index that describes the areas of accumulation of surface runoff and soil moisture, influencing the decomposition of organic matter.
32. **GHG Inventory**: Accounting for GHG emissions and/or removals of an organization, project or product.
33. **Minimum Soil Tillage**: Soil tillage technique that **minimizes soil disturbance**, limiting the working depth.

34. **Leakage (Losses)**: Net change in greenhouse gas emissions outside the project boundaries, which need to be quantified and mitigated.
35. **Litter**: Non-living biomass (size > 2mm, < 10cm) in various states of decomposition, lying on or within mineral or organic soil.
36. **IPCC Tiers:** Three levels of emissions estimation with increasing data requirements and analytical complexity for greater accuracy: Level 1 (predefined factors), Level 2 (country-specific data), Level 3 (methods, models and repeated measurements over time).
37. **Guarantee Level**: Level of assurance on the GHG declaration.
38. **Metodo Walkley & Black**: Standard method for estimating the percentage of organic carbon in soil.
39. **Best Agricultural Practices (BAP):** Agricultural management techniques that promote environmental, economic, and social sustainability.
40. **Monoculture**: Cultivation system that involves the cultivation of **a single plant species** in one field at a time.
41. **Monitoring:** Continuous or periodic assessment of GHG emissions, GHG removals, or other related data. It involves the systematic collection of data related to project activities, carbon stocks, and land use changes.
42. **Monitoring, Reporting and Verification (MRV)**: Continuous system that includes project data collection, periodic reporting, and independent verification to ensure compliance with standards and the reliability of declared results.
43. **Notarization**: Process that ensures the authenticity, integrity, and traceability of carbon credits, using secure and transparent digital ledgers, often using blockchain technology.
44. **Involved Party**: A person or organization that can influence, be influenced by, or perceive itself as being influenced by a decision or activity.
45. **Stay**: The need to ensure that the **climate benefits are maintained over time** It focuses on the need to preserve sequestered carbon in the long term.
46. **Pesticides**: Chemicals used to control organisms harmful to crops.
47. **Carbon Pool**: A system that can store and accumulate carbon. Examples include wood, soil, roots, litter, and deadwood.
48. **Global Warming Potential (GWP)**: Index based on the radiative properties of GHGs, which measures the radiant strength of a unit mass of a given GHG relative to that of a unit mass of carbon dioxide over a specific period of time.
49. **Greenhouse Gas Program (GHG Program)**: Voluntary or mandatory international, national or subnational system or scheme that records, accounts for or manages emissions, removals, emission reductions or increases in removals of GHGs outside the organization or project.
50. **Greenhouse Gas Project (GHG Project)**: One or more activities that alter conditions in a GHG baseline resulting in reductions in GHG emissions or increases in GHG removals.
51. **Greenhouse Gas Project Proposer (GHG Project Proposer)**: Individual or organization that has overall control and responsibility for a GHG-related project.
52. **Greenhouse Gas Report (GHG Report):** A stand-alone document intended to communicate information about an organization's or project's GHG emissions to its intended users. Reporting consists of providing information on emissions and removal estimates, estimation methods, procedures, and systems.

53. **Record:** Database that manages the issuance, trading, and retirement of carbon credits in carbon markets. The carbon farming registry is public and online, and contains information on carbon removal units generated, available, and sold.**Immutable Register**It is a digital ledger on blockchain where transactions cannot be modified or deleted.
54. **Greenhouse Gas Removal (GHG Removal)**: Removal of a GHG from the atmosphere.
55. **Net GHG Reduction**: Occurs when an agricultural or land management activity **reduces emissions** of GHG compared to the reference situation (baseline).
56. **Net GHG Removal**: It occurs when an agricultural or land management activity leads to a **major seizure** of GHGs from the atmosphere compared to the baseline situation.
57. **Baseline Control Site**: Defined area managed according to pre-project practices (baseline) for direct measurement of change in baseline soil organic carbon (SOC) stock.
58. **Reference Scenario**: Hypothetical reference case that best represents the most likely conditions in the absence of a proposed GHG project.
59. **Greenhouse Gas Reservoir (GHG Reservoir)**: A component, other than the atmosphere, capable of accumulating, retaining, and releasing GHGs. An element that contributes to GHG emissions and/or removals.
60. **Hedges, Rows and Integrated Woods**: Integration of trees and shrubs into agricultural systems to provide various ecosystem services, including carbon sequestration.
61. **Greenhouse Gas Information System (GHG Information System)**: Policies, processes and procedures to establish, manage, maintain and record information relating to GHGs.
62. **Smart Contracts**Self-executing contracts encoded on blockchain, which allow for the automation of operations based on predefined conditions, improving efficiency and trust.
63. **Greenhouse Gas Source (GHG Source)**: Process that **releases a GHG into the atmosphere**. An element that contributes to GHG emissions.
64. **GHG Sources, Sink and Reservoirs (SSR)**: Collective term for elements that contribute to GHG emissions and/or removals. Sources release GHGs, sinks remove them, and sinks accumulate them. Includes SSRs controlled by the developer, related to the project by material or energy flows, and influenced by the project.
65. **Natural Inorganic Substances**: Substances of mineral origin that can be used as soil improvers or fertilizers, which have not undergone chemical synthesis processes.
66. **Stratification**: Process of dividing the project area into homogeneous zones based on variables such as topographic moisture index (TWI), normalized difference vegetation index (NDVI), and soil type, to optimize sampling.
67. **Remote Sensing**: Method for determining biomass and monitoring agricultural practices and land cover changes using **satellite images** and aerial data, often combined with field measurements (ground truthing).
68. **TC (Total Carbon)**: Total carbon content.
69. **TCR Carbon Standard**: Structured set of guidelines and procedures developed by Carborea to ensure the governance, implementation and certification of carbon credit projects in the context of the Voluntary Carbon Market (VCM).

70. **TCR (Trusted Carbon Reduction)**Carborea's methodology was developed to provide a comprehensive and rigorous framework for quantifying, monitoring, and reporting greenhouse gas (GHG) emissions and removals in agricultural contexts.
71. **TIC (Total Inorganic Carbon)**Total inorganic carbon tends to prevail in soils developed from coarse sediments.
72. **TOC (Total Organic Carbon)**: Total organic carbon tends to prevail in soils developed from fine-grained sediments.
73. **Sample Unit**: A defined area within the project selected for measurement and monitoring, which must be homogeneous in terms of management activities, soil type, climate, etc.
74. **Intended User**: An individual or organization identified as relying on GHG information to make decisions.
75. **Validatore**: Competent and independent person responsible for performing and reporting a validation.
76. **Validation**: Process for evaluating the **reasonableness of assumptions, limitations and methods** which support a statement on the outcome of future activities. This occurs before the project is implemented.
77. **Verifier**: Competent and independent person responsible for carrying out and reporting an audit.
78. **Verify**: Process for evaluating a statement with **historical data and information** to determine whether the declaration is factually accurate and compliant with the criteria. This occurs after the project has been implemented.

PRINCIPLES

In the context of the verification and validation of greenhouse gas (GHG) reduction and removal projects in the agricultural sector, the**TCR (Trusted Carbon Reduction)** It is based on core principles that guarantee its integrity and reliability. These include accuracy, consistency, completeness, conservatism, and transparency, essential to the credibility of the carbon credits generated, specifically:

- **Accuracy:** This principle requires that quantifications of GHG emissions and removals be **precise and reliable**, with the aim of minimizing errors and uncertainties. Accuracy is achieved through the application of consolidated methodologies such as the IPCC Guidelines and recognized standards (e.g. Verra VM0042, CDM AR-AMS0007), the collection of **reliable and rigorous data**, and the continuous assessment of inherent data uncertainty. Greater detail and the use of higher-tier approaches (Tier 2 or Tier 3) in IPCC methodologies are encouraged to improve accuracy and reduce uncertainty in inventories.
- **Consistency**: Consistency ensures that data and methodologies are **applied evenly over time**, allowing for direct comparison of environmental performance across time periods. This requires consistent use of definitions, soil classifications, and management categories. Even when modifying parameters or using country-specific data, it is good practice to document and explain deviations from the IPCC default data to maintain consistency.
- **Completeness:** The principle of completeness requires that the GHG inventory **include all relevant GHG sources, sinks and reservoirs** within the project boundaries, in addition to all managed land areas and non-CO_2 greenhouse gases, where applicable. It is essential to avoid omissions or double counting when representing land areas and activities. The TCR Monitoring, Reporting, and Verification (MRV) system is designed to ensure this comprehensive coverage.
- **Conservatism**: This principle imposes a **cautious approach** in estimating GHG reductions or removals to avoid overestimating climate benefits. For example, for the baseline, assumptions and projections are selected conservatively, and in some cases, CO_2 emissions from soil organic carbon losses or fossil fuels may be assumed to be zero to maintain a conservative approach in the standardized baseline. The TCR specifies that, in the event of discretion in choosing a value for a parameter, the principle of conservatism must be applied.
- **Transparency:** Transparency is essential to building trust and credibility in the carbon market. It requires **clear, complete and accessible documentation** of all information, including the methodologies used, the data collected, and the results. The TCR actively promotes transparency by publishing monitoring reports in a public ledger and using blockchain technology to notarize carbon credits, ensuring the immutability and verifiability of transactions.

INTRODUCTION TO GHG PROJECTS

The development cycle of a greenhouse gas (GHG) reduction and removal project in agriculture, as outlined by the methodology**TCR (Trusted Carbon Reduction)**, is divided into sequential and interdependent phases: the definition of the **baseline**, the initial **validation** and the periodic **verification**. These steps are crucial to ensuring the integrity, credibility, and traceability of the carbon credits generated.

- **Baseline (Reference Scenario):** The baseline, or "business-as-usual" (BAU) scenario, represents the **quantitative reference of GHG emissions and/or removals that would have occurred in the absence of the project.** In agriculture, it is the expected carbon removal performance in the absence of carbon farming practices. This scenario is constructed based on historical data from conventional farming practices, industry trends, and information from reliable sources. For a conservative approach, CO2 emissions from soil organic carbon (SOC) losses in cropland can be assumed to be zero. Baseline estimates can be modeled or measured directly at "baseline control sites."
- **Validation (ex-ante):** Validation is the process of **evaluating the reasonableness of assumptions, constraints, and methods that support a statement about the outcome of future activities.** It happens **only once, before the project implementation.** During this phase, an independent Validation and Verification Body (VVB) examines several crucial aspects:
  - **Project design**: Ensures that objectives, scope, location and duration are clearly defined and realistic.
  - **Baseline scenario**: Verify that the baseline is credible, accurate, representative and that the assumptions are conservative.
  - **Additionality**: Confirm that the GHG reductions/removals generated by the project are additional to the BAU scenario.
  - **Monitoring plan**: Assess that it is robust and includes procedures for measuring, collecting and analyzing relevant data.
  - **Stay**Ensure that the project includes measures to ensure climate benefits over time, such as a buffer system (for example, the TCR adopts a 10% buffer of credits). This buffer is justified by a risk assessment based on the Verra "AFOLU Non-Permanence Risk Tool."
  - **Compliance**: Verify alignment with international standards such as ISO 14064-2, IPCC Guidelines, Verra VM0042 and CDM AR-AMS0007.
- **Verify (ex-post)**: Verification is the process of **evaluation of a statement with historical data and information to determine whether the statement is materially correct and compliant with criteria.** It happens **periodically after project implementation.** The VVB conducts these assessments for:
  - **Assess actual GHG reductions/removals:** Comparing project results with the baseline scenario and monitoring reports. This includes measuring carbon sinks (SOC, living biomass, harvested wood products) at two specific time points (t0 and t1).
  - **Conduct on-site inspections:** To verify the implementation of sustainable agricultural practices and field data collection.
  - **Review monitoring data:** Evaluating their completeness, accuracy, and consistency. SOC measurements are required at least every five years or

prior to each verification event where there is an interest in quantifying GHG emissions/reductions from BAPs for the purpose of generating carbon credits.
  - **Verify data quality management**: Ensuring that QA/QC (Quality Assurance/Quality Control) procedures have been implemented.
  - **Confirm your stay**: Evaluating the maintenance of the buffer system and risk mitigation measures.
  - **Review the reporting**: Ensuring that monitoring reports comply with standards and that information is clear, complete and transparent.

The project development cycle is a continuous process of planning, implementation, measurement, and review. The TCR adopts a continuous approach to **MRV (Monitoring, Reporting and Verification)** which includes systematic data collection, periodic reporting, and independent verification. The **blockchain** technology is used for the **notarization** of carbon credits, ensuring security, transparency and immutability of the digital ledger.

BASELINE MEASUREMENT

The process of **project baseline measurement** is a series of systematic and rigorous actions to be undertaken to establish a baseline scenario ("business-as-usual" or BAU). This hypothetical scenario represents the greenhouse gas (GHG) emissions and/or removals that would have occurred in the absence of the project activities. This quantification is essential for determining the additional climate benefits generated by the project, such as those predicted by the TCR (Trusted Carbon Reduction) methodology.

To measure the baseline, you need to take the following actions:

1. **Define and characterize the historical reference scenario:** It is crucial to establish a historical reference period, which can extend up to a **maximum of three years.** During this period of time, you must collect **detailed historical data on conventional farming practices** prevalent in the area, which typically include continuous cropping systems, monoculture, bare-soil plowing (with a plow that inverts the soil layers), crop residue removal (often by burning), and the application of inorganic nitrogen fertilizers. The baseline must be credible, accurate, and representative of the BAU situation.

2. **Quantifying net GHG removals in the baseline (CRbaseline):** This step evaluates the carbon sequestration potential in the BAU scenario. For a **conservative approach**, CO_2 emissions from soil organic carbon (SOC) losses in baseline cropland are often **considered equal to zero**, as the literature suggests that conventionally managed soils act as net sources of CO_2. For other carbon sinks, calculations or measurements may be necessary:

    - **Soil Organic Carbon (SOC) Measurement:** Although baseline SOC losses are often assumed to be zero for conservative purposes, the project can include an initial estimate (time t=0) coinciding with reforestation activity or the start of new carbon farming practices or BAPs, and subsequent periodic surveys where this data is to be considered for the purpose of generating additional carbon credits. These can be achieved through direct measurements in "**baseline control sites**", i.e., areas managed according to pre-project practices. Soil samples must be taken from a minimum depth of 30 cm and analyzed by accredited laboratories, using standard methods such as that of Walkley & Black. The amount of SOC is calculated using the formula: $SOC = BD \times C\% \times D \times (1 - F)$ Where:
        - BD is the apparent density of the soil (Mg/m^3 or g/cm^3)
        - $C\%$ is the percentage of organic carbon in the soil (%)
        - D is the soil sampling depth (m)
        - F is the volume fraction of coarse fragments (> 2 mm)
    - **Plant measurement (living biomass)**: The inclusion of this reservoir is necessary if the project activities result in a significant increase. For trees, quantification is based on measuring the height and **diameter at breast height (DBH)** at 1.37 m above ground level. Aboveground biomass (AGB) is

estimated using allometric equations and a Biomass Expansion Factor (BEF):$AGB = BEF \times V \times \rho$ Where:

- In is the volume of the tree, calculated as$V = BA \times H \times F$(BA = Basal Area, H = Height, F = Shape Factor)
- BEF is the Biomass Expansion Factor
- R is the density of wood (g/cm³) The below ground biomass (BGB) can be estimated from the AGB using a **roots-to-topsoil ratio (R)**: $BGB = AGB \times R$The total carbon stock (CS) in biomass is given by:$CSbiomassa = (AGB + BGB) \times CF$ Where:
- CF is the carbon fraction (usually 0.47 for dry biomass) The change in carbon stock (ΔC) over time is calculated as:$\Delta CSOC,LB,HWP = (Ct1 - Ct0) / (t1-t0)$And converted to CO2 equivalent with:$\Delta CO2 = -44/12 \times \Delta C$

- **Harvested Wood Products (HWP):** Although not always foreseen in agricultural TCR projects, this reservoir can be included and monitored for long-term storage. Estimates can follow IPCC guidelines or specific methodologies such as VM0005-10. **Tier 1 methods**The IPCC guidelines (Volume 4, Chapter 12) for harvested wood products suggest the use of default values for all activity data and parameters.

3. **Calculate the GHG emissions that would occur in the baseline scenario (GHGbsl):**This includes direct and indirect emissions from conventional agricultural practices. Total baseline emissions ($GHGcf;bsl$) are calculated with the following formulas:

- $GHGcf;bsl = GHG(INF) + GHG(FUEL) + GHG(OA) + GHG(CC)$ Where:
- $GHG(INF)$: emissions from inorganic nitrogen fertilizers$GHG(INF) = X(INF) \times EF(INF) / 1000$ (X= amount of nitrogen applied,IF= emission factor)
- $GHG(FUEL)$: emissions from fossil fuel consumption for agricultural operations$GHG(FUEL) = X(FUEL) \times EF(FUEL) / 1000$ (X= amount of fuel consumed,IF= emission factor)
- $GHG(OA)$: emissions from organic nitrogen fertilizers$GHG(OA) = X(OA) \times EF(OA) / 1000$
- $GHG(CC)$: emissions from nitrogen-fixing cover crops$GHG(CC) = X(CC) \times EF(CC) / 1000$: Emissions such as soil methanogenesis or enteric fermentation from livestock are generally excluded from the baseline under these methodologies unless they are widespread or significantly impacted by the project. All calculations must be expressed in **tons of CO2 equivalent (tCO2e)**, using Global Warming Potential (GWP) values (e.g. 100 years after the IPCC Fifth Assessment Report).

The project baseline validation report shall focus on the **rigorous and systematic evaluation of the baseline scenario**, which represents the greenhouse gas (GHG) emissions and/or removals that would have occurred in the absence of the project activities.

The **Validation and Verification Body (VVB)**, as an independent third-party body, carries out the validation phase (ex-ante). During this phase, the VVB must:

- **Examine the baseline scenario**, ensuring that the baseline is **credible, accurate and representative of the "business-as-usual" (BAU) situation** without the intervention of the project.
- Check that the **assumptions and projections are conservative**.
- Evaluate the **baseline justification**, which is based on the diffusion of conventional agricultural practices in the reference sector, supported by analysis of historical data and sector trends. The description of the baseline must include the**prevalent conventional agricultural practices**, such as continuous cultivation, monoculture, bare soil ploughing with a ploughshare, removal of crop residues and the application of inorganic nitrogen fertilizers.
- Confirm that the baseline for agricultural management improvement (ALM) interventions is calculated on a **minimum historical base of five years preceding the start of the project**, or a minimum of three years including at least one complete crop rotation.

The **project developer** is the entity responsible for the **definition and quantification of the baseline.** The document supporting the project data (Project Design Document) must therefore include:

- The **detailed description of the baseline**, including cultivation types and conventional management practices.
- The **quantification of net GHG removals in the baseline (CRbaseline)**, often assumed to be zero for soil organic carbon (SOC) losses in cropland for a conservation approach, as conventionally managed soils act as net sources of CO2. However, the project requires an initial estimate (t=0) of SOC, which can be achieved through direct measurements at "baseline control sites."
- The **quantification of GHG emissions in the baseline scenario (GHGbsl)**, which includes direct and indirect emissions from inorganic nitrogen fertilizers, fossil fuel consumption for agricultural operations, organic nitrogen fertilizers, and nitrogen-fixing cover crops.
- L'**identification of relevant carbon sinks**(e.g. aboveground and belowground biomass, SOC, harvested wood products), and associated greenhouse gases (CO2, CH4, N2O), with justification for inclusion or exclusion.
- An **uncertainty assessment** associated with the baseline estimate, considering the input data, emission factors and quantification methodologies used.
- The **methodologies and standards used** for baseline quantification, such as the IPCC Guidelines for National Greenhouse Gas Inventories, Verra VM0042 and CDM AR-AMS0007.

- The **monitoring plan** and data quality management (QA/QC) procedures to ensure the accuracy and traceability of information.
- The **additionality**, including a section highlighting the benefits to project management resulting from the issuance of carbon credits.

ANNUAL PROJECT MEASUREMENT

Annual project measurements are used to determine the net carbon removal benefit compared to the baseline. The basic formula used for calculation is: **Net carbon removal benefit = CRbaseline – CRtotal – GHGincrease** where the GHGincrease value is always considered positive if non-zero. This formula allows us to quantify the CO2 removals generated by the project's activities as the difference between the project scenario (with virtuous practices) and the standardized baseline. All emissions and removals are expressed in tons of CO2 equivalent (tCO2e). The calculation of the credits generated by the change in SOC must be performed by comparing the measurements at time t with those at time t-1.

During the onboarding phase, the condition of the land must be determined by submitting the relevant documentation (e.g., delivery notes, invoices, certifications). For this purpose, the most relevant document is the notification of uprooting sent to the regional government. Optionally, the project developer will provide notification of replanting when new trees are planted.

Subsequent inspections must include the presentation of the documentation necessary to verify the status of seizure/absorption of the project (e.g. satellite and/or aerial photos, remote sensing data, etc.)

The GHG increase resulting from the uprooting activity must be considered in the baseline calculation only if the activity was conducted by the project developer.

- **CRbaseline (Carbon Removal Relative to Standardized Baseline):** The baseline represents the GHG emissions and/or removals that would occur in the absence of the project intervention, based on prevailing conventional agricultural practices. For field crops, it includes continuous cropping systems, monoculture, bare soil plowing, use of plowshares, residue removal, and application of inorganic nitrogen fertilizers. For a conservative approach,**CO2 emissions from soil organic carbon (SOC) losses in cropland in the baseline are often assumed to be zero.** The baseline for agricultural management improvement (ALM) interventions must be calculated on a historical basis of at least five years preceding the start of the project.
- **CRtotal (Total carbon removal from project activities)**: This value is calculated based on the **measuring carbon sinks at two points in time** (t0 at the beginning of the certification period and t1 at the time of measurement) to evaluate the storage variations. The carbon sinks included are the **soil organic carbon (SOC)**, the **living biomass (LB)**(above and below ground) and the **harvested wood products (HWP)**The annual change in carbon stocks in a reservoir is calculated as`(Ct1 - Ct0) / (t1-t0)`.
    - The measurements of the**SOC**are required at regular intervals, typically **every 3-5 years or in any case before each verification event.** Soil samples are taken from a minimum depth of 30 cm and analyzed by accredited laboratories using standard methods such as that of Walkley & Black. SOC is calculated by multiplying the soil bulk density (BD) by the percentage of organic carbon (C%) and the sampling depth (D), taking into account the fraction of coarse fragments.

- During the **first check**(and therefore the first year of carbon credit generation) is included in the absorption of **all organic matter** of the plant from seed. The requirement is that the plant must have been grown from seed or purchased, in which case it must be less than 2-3 years old to do so.
- For the **living biomass** (aboveground and belowground), measurements are taken to estimate woody biomass (e.g., diameter at breast height (DBH), tree height) and herbaceous biomass. For afforestation/reforestation (A/R) and sustainable forest management (SFM) projects involving harvesting, a commitment to continue management practices that maintain or increase carbon stocks must be demonstrated. Belowground biomass can be calculated as a fraction of aboveground biomass using the Root-to-Shoot Ratio (RTR). Carbon conversion to CO2 is obtained by multiplying the change in carbon stock by the conversion factor. `(-44/12)`.
- The CDM AR-AMS0007 v3.1 methodology is integrated to quantify emissions and removals from afforestation/reforestation activities. Verra VM0042 v2.0 is used to quantify soil carbon sequestration. The IPCC guidelines provide the framework for quantifying emissions.

- **GHGincrease (Increase in GHG emissions in the project):** This component considers the increase in direct and indirect GHG emissions due to project implementation, compared to the baseline. Only positive differences are included. Emissions considered include those from inorganic and organic nitrogen fertilizers (GHG(INF), GHG(OA)), fossil fuel consumption for agricultural operations (GHG(FUEL)), and cultivation of nitrogen-fixing cover crops (GHG(CC)). CO2 emissions from fossil fuels are often conservatively excluded if project practices reduce them. Soil methanogenesis and enteric fermentation are generally excluded due to their negligible impact in these agricultural settings.

The generation of an **annual report:** It is a fundamental step in monitoring and reporting carbon farming projects, ensuring transparency, traceability, accuracy, comparability, completeness, and consistency of GHG removal benefits. While not all physical measurements of carbon sinks, such as soil organic carbon (SOC), are performed annually (typically every 3-5 years or before each audit), internal monitoring is conducted on an **annual basis.** At the end of each certification year, net carbon removal data is entered into the carbon farming registry.

The **project proponent** is responsible for compiling and submitting these reports, while a**Certification Body (CB)**The accredited body is responsible for preparing and publishing them in the public registry. These detailed project progress reports include GHG reductions, carbon sequestration, and the adoption of sustainable agricultural practices. The core of the annual report is the **quantification of the net benefit of carbon removal**, calculated using the formula:`Net benefit = CRbaseline − CRtotal − GHGincrease`The report must detail changes in CO2 sequestration, changes in CO2 emissions, any leakage, and the recalculation of the permanence buffer. Key certification information, such as net removal benefits, removals compared to baseline, total removals, and the increase in GHG emissions, must be clearly confirmed.

To ensure data reliability, procedures are implemented**Quality Assurance and Control (QA/QC)**, with regular calibration of measuring equipment and careful independent review of data to prevent or correct errors and omissions. The data used for the calculations come from a multilevel approach that combines direct information from farms (questionnaires, interviews, farm records) with information from secondary sources (databases, literature) and remote sensing technologies (e.g., satellite imagery). All data and summarized results must be made available on the project website and in the public registry, often based on blockchain technology, to ensure maximum **transparency and traceability** throughout the life cycle of carbon credits.

CONFORMITY ASSESSMENT

Conformity assessment is used in the market to increase confidence in the conformity of an item.

- **Initial validation:** If a project has already been launched, validation can be performed retroactively within a maximum of five years of project start-up. Should this occur, to ensure greater project transparency, the operator will only be able to generate carbon credits retroactively for a maximum of five years of project activity.
- **Checks**: Actual data audits must be performed annually to monitor the effectiveness of the actions undertaken and evaluate any adjustments, with the first audit to be conducted on-site followed by a remote audit the following year, alternating between the two types of audit.

Resource requirements
The audit team must have adequate and diversified skills to evaluate climatic aspects and environmental management techniques.

Validation/Verification Program
The initial validation (year 0) ensures compliance of the methodologies and must be carried out on-site, with a subsequent on-site audit at the end of the year. Audits will then continue annually, alternating remote and on-site audits, in order to monitor the credits generated and the improvements achieved.

Validation/verification processes
This paragraph describes the specific requirements for the validation and verification process of this Internal Practice.
All elements required by ISO/IEC 17029 must be respected.

Pre-assignment
In the pre-assignment phase, the following information must be collected at least:
- organization personal data;
- documents to be verified and the scope of the evaluation
- supporting data and information
- any first-party/second-party/third-party audits.

Assignment

The validation/verification body stipulates a**formalized agreement**with each customer for the provision of validation/verification activities. This agreement ensures compliance with the

The agreement stipulated must include clauses that commit the customer to:

- Respect the **validation/verification requirements**.
- Provide all the **necessary provisions** to carry out activities, including reviewing documentation and accessing relevant processes, areas, records, and personnel.

Performing validation/verification

In particular, for the evidence collection activity, the Body must ensure traceability, through the data management process and any other additional analysis or calculation, and identify inaccuracies considering their relative materiality.

Evidence gathering includes document review and verification.

For the following such practice is required a reasonable level of assurance and materiality, which therefore reduces the risk of the assertion being untrue to a low level and which allows the Body to issue a positive opinion.

ANNEX 1 - ACTIONS TO BE TAKEN

Requirements and Timeline for Validation and Verification of GHG Projects

A validation requirement for a project to be accredited is that the age of the trees at the time of planting is no more than 5 years.

Subsequent checks will be carried out following the chronology indicated for each individual type (remote, on-site) and the age of the tree.

- Example 1: At T0 the planting has yet to take place
    - - Activity: validation
    - - Years credited: 0
    - - First check: after planting (remote sensing + on-site)
    - - Subsequent on-site inspection: after 5 years
    - - Subsequent checks: annual remote sensing, on-site every 5 years - tree age

- Example 2: At T0 the age of the trees is 2.5 years.
    - - Activities: validation + verification (remote sensing + on-site)
    - - Years credited: 2.
    - - Subsequent on-site inspection after 2.5 years.
    - - Subsequent checks: annual remote sensing, on-site every 5 years - tree age

Validation

Validation is a process of evaluating the **reasonableness of assumptions, limitations and methods** which support a statement about the outcome of future activities. It is performed **once, before the project implementation**.

Check	Responsible	Frequency	Compliance Criteria	Execution Mode	Communication Methods
1.1 Project Design, Reference Scenario and Monitoring Plan	**Third party validator**(Certification Body)	Once, **before project implementation**	The project design, baseline scenario and monitoring plan must be **realistic, credible and compliant with the ISO 14064-2 standard**.	The third-party validator evaluates the project design, baseline scenario, and monitoring plan proposed by the proponent.	The third party validator provides**detailed reports**which outline the results of the evaluations. This information must be included in the**Project Design Document (PDD)**.
1.2 Project Suitability (Site and Land)	**Project Proposer**	Once, **before project implementation**	The site must be **compatible** with the project objectives; the must be verified **ownership and legal availability** of the land; the **nature and state of the soil** (e.g. degraded, abandoned or deforested lands); the following must be considered:**local ecological conditions** and the **compliance with current regulations**.	The proponent carries out a preliminary assessment of the site's suitability.	The project location and relevant eligibility information must be included in the PDD. A**file KML**(Keyhole Markup Language) of the spatial boundary of the project, with a clear description and metadata.

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1.3 Additionality	**Project Proposer**	Once, **before project implementation**	The climate benefits of the project must be **actually additional** compared to a "business-as-usual" (BAU) scenario. The activity must not be tied to mandatory legal requirements. Additionality must be demonstrated through **chemical and physical tests**, with monitoring of chemical substances (UNI EN ISO/IEC 17025 accredited laboratory analyses) and chemical-physical parameters (e.g. satellite simulation models). The financial additionality must demonstrate an **actual advantage** derived exclusively from the sale of credits. The baseline for management improvement interventions must be calculated on a **minimum historical base of 2 years** preceding the project.	Analysis of historical agricultural data and GHG emissions records. Use of comparative studies. Consultation with local farmers, agricultural experts, and regional authorities. The proponent must provide convincing evidence.	The additionality assessment must be included in the PDD. The online public registry must include information on certified projects that generate carbon credits.
1.4 Permanence and Buffer	**Project Proposer**	Once, **before project implementation**	The project must ensure the **long-term stay** of sequestered carbon or emissions reduction. It must include a**minimum 10% credit buffer**(or a lower value with a carefully documented risk mitigation strategy). Carborea requires a minimum commitment of **20 years** to maintain green assets and offsets.	The proponent must explicitly specify the number of years for which they intend to maintain the green assets and carbon offsets. Implement preventative and corrective measures to reduce the risk of carbon inversions.	The permanence assessment must be included in the PDD and the TCR methodology details the calculations.

1.5 Risk Assessment	**Project Proposer**	Once, **before project implementation**	The **overall risk scor e**must be less than **60 points**; the risks **interiors** lower than **35**, **exteriors** lower than **20**, **natural** lower than **35.** The project must have an **adaptive management plan.** The management team must include individuals with **specific skills.** There must be a **complete training plan** for all participating farmers.	The Verra tool is applied in "**AFOLU Non-Permanence Risk Tool**" Version 4.2 to assess risk categories (internal, external and natural).	The results of the risk assessment must be documented in the PDD.
1.6 Absence of Intentional Deforestation and Invasive Species/Previous Offsets	**Project Proposer**	Once, **before project implementation**	The project site must not have undergone **deliberate deforestation** within five years prior to the start of the project or additional activities. They must be used **exclusively non-invasive species** (preferably indigenous). The project **must not have already created or received other compensation units** of carbon for the same activities.	Carborea conducts an audit using historical satellite data. The proponent presents proof of tree purchase and species details.	Details regarding the species planted and the absence of previous offsets must be provided in the Carborea Onboarding Form.
1.7 Data Quality and Uncertainty	**Project Proposer**	Once, **before project implementation**	Procedures must be established and implemented in data **quality management** (QA/QC) suitable for managing information, including that related to the assessment of the degree of uncertainty. The uncertainty associated with the baseline estimate must take into account the uncertainty of the input data, emission factors, and quantification methodologies used.	Quality management (QA/QC) procedures must be established and implemented. Standard uncertainty propagation techniques must be applied.	Project documentation must include data quality management. The GHG report must include a statement on the degree of uncertainty.

1.8 Compliance with Standards and Methodologies	**Project Proposer**	Once, **before project implementation**	Compliance with**ISO 14064-2**, **IPCC Guidelines**, **Verra VM0042**, **CDM's AR-AMS0007 v3.1**, **LIFE C-Farms**.	The applicant must justify the choice of recognized origins if there is more than one, and document any deviations from the recognized criteria and procedures.	The PDD must specify the methodologies and standards used as reference.
1.9 Ownership of Credits	**Project Proposer**	Once, **before project implementation**	Ownership of credits must be **defined in advance and unambiguously** for the certification period (5 years). If the proponent is not the owner of the area, a **written consent form** of the owner who authorizes the intervention and is aware of the restriction.	Contractual definition of ownership of credits.	Written contractual documentation demonstrating compliance with applicable laws must be provided.
1.10 Methodological Specifications	**Project Proposer**	Once, **before project implementation**	**Organic Farming:**Project initiated after notification of biological activity or with documentation attesting the specific purpose of generating credits prior to certification. Inclusion/exclusion of carbon pools (significant >10%, non-significant <5%) must be justified.	Submission of appropriate documentation certifying compliance with specific methodologies. Identification of carbon pools.	The PDD must detail compliance with all applicable methodological specifications.

Verify

Verification is a process of**evaluation of a statement with historical data and information**to determine whether the declaration is materially correct and compliant with the criteria. It is carried out**periodically after project implementation**.

Check	Responsible	Frequency	Compliance Criteria	Execution Mode	Communication Methods
2.1 Actual GHG Reductions and Removals	**Third party verifier**(Certification Body)	Annually. Remote monitoring annually.	Actual GHG emission reductions and carbon sequestration must be **compliant** to the project's baseline scenario and monitoring reports. Quantification must be **accurate and credible.** The calculation of the*CRtotal*It is based on the measurement of carbon sinks (SOC, living biomass, harvested wood products) at two points in time (t0, t1).	**On-site and remote audit with review of monitoring data**. **Soil sampling and analysis**for Soil Organic Carbon (SOC). Use of **soil carbon models**Measurement of carbon stocks in trees and shrubs before each on-site survey. Analysis of aerial and satellite data with **ML and AI** to detect land use changes and estimate biomass.	The third party verifier provides **detailed audit reports**, including findings, discrepancies, and recommendations. **Monitoring reports** must be published in the **public register.** The summarized data and results are available on the project website.
2.2 Implementation of Sustainable Agricultural Practices	**Third party verifier**(Certification Body)	Continuous internal monitoring**annual** Annual audit.	Sustainable agricultural practices must be **actually implemented** by participating farmers. The area occupied by recognized carbon removal uses must not decrease. The project must progress toward its stated objectives.	On-site inspections. Data collection through surveys, direct observations, and agricultural records. Monitoring through**remote sensing**for vegetative health, land cover changes and biomass accumulation.	Monitoring reports must document the implementation of practices.

2.3 Data Quality and Control (QA/QC)	**Project Proposer** (Carborea S.B. S.R.L.), **Third party verifier**	SOC measurements shall be conducted at least **every 5 years** or before each verification event if more frequent. Emission factors must be monitored **every 5 years**.	Procedures must be applied**QA/QC**in accordance with the monitoring plan. The measuring and monitoring equipment must be**calibrated, verified and maintained**Errors and omissions in the data must be prevented or corrected.	Training of staff and participants on data collection and reporting procedures. Regular calibration of monitoring equipment. Independent review and verification of data. Internal data controls (input, transformation, output). Use of internal protocols and accredited laboratories for soil sample analysis.	The collected data and the summarized results must be available on the project website to ensure **transparency**.
2.4 Registry Management and Blockchain Transparency	**Carborea S.B. S.R.L.**	Continuous	The carbon farming register must be **public and available online**, reporting information on the removal units generated, available and sold, tracking issued certificates, projects and buyers. The technology **blockchain ensures security and transparency** of the data.	Recording carbon credits on a **secure and transparent digital register** (blockchain). **Notarization on blockchain** to ensure the immutability and verifiability of information. Use of**Smart Contracts**for automated verification and immutable records.	The monitoring report must be published in the **public register.** The information contained in the register and project reports must be available for public consultation. Aerial and satellite data are transformed into **interactive maps and visualizations** for stakeholders.
2.5 Periodic reporting (General)	**Carborea S.B. S.R.L.**	**Annually**	Completing and sending **detailed reports** on project progress. Reports must comply with reference standards and be made public for verification purposes. Any**deviation**from the certification project must be**reported**.	Compiling annual reports including data on GHG emission reductions, carbon sequestration, and practice adoption. Monitoring data analysis.	Reports must be shared with stakeholders (funders, participating farmers, public).

2.6 Specific Quantification (e.g. Organic Farming)	**Project Proposer**	**Annual**(for monitoring)	Stock quantification must be based on annual monitoring or scientific studies. Mandatory carbon stock variables must be collected (species, shell biomass, soil, water, bottom mud, PAR/LAI/NDVI, gaseous CO2). Annual CO2 emissions for management (fuel, electricity, other processes, product emissions) must be quantified.	Surveys on elemental samples. Minimum sampling depth of 30 cm for soil/bottom mud. Sampling and measurement methodologies must be statistically appropriate and compliant with the IPCC Guidelines.	Drafting of an **explanatory manual of the counts.** Presentation of summary indexes of fixation activities with related documentation.

ANNEX 3 - CALCULATION METHODOLOGIES

1. Massa Soprasuolo (AGB - Above Ground Biomass)

The AGB is estimated using allometric equations based on**DBH (diameter at breast height) and tree height**:

BCEF = BEF×ρ (*Source IPCC - p.*)
AGB = BCEF×V (*Source IPCC - p.*)

Where:

- **In**= Volume of the tree, calculated as:
  H=W×H

with

  - **NOT**= Area basale (π×(DBH/2)2)

- **H**= Total height of the tree

- **BEF** = Biomass Expansion Factor (*Source: IPCC 1996 - Table 4.5*)
- **BCEF** = Biomass Conversion and Expansion Factor (*Source: IPCC 2006 - Table 4.5*)
- **r**= Wood density (g/cm³, available in literature for specific species)

Parameters required for the calculation

- **DBH**(breast-width diameter, cm)
- **H**(total height of the tree, m)
- **Wood density (ρ\rho)**
- **BEF (optional, if biomass expansion method used)**

The 2006 IPCC Guidelines for National GHG Inventories, specifically volume 4 "Agriculture, Forestry and Other Land Uses (AFOLU)" in chapter 4.5, provide reference values for the determination of parameters such as carbon fraction (table 4.3), biomass expansion factor (table 4.5), wood density (table 4.14) and root-to-stand ratio (table 4.4).

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2. Massa Sottosuolo (BGB - Below Ground Biomass)

Root biomass can be calculated as a fraction of AGB using the**Root-to-Shoot Ratio**.

$BGB=AGB\times R$

Where:

- **R**= Root-to-Shoot Ratio (usually 0.26 for tropical forests, variable for other species)

Parameters required for the calculation

- **AGB** (see above)
- **Root-to-Shoot Ratio (RR)**

3. Soil Organic Carbon (SOC)

SOC is calculated at a specific depth (e.g. 0-30 cm) with the formula:

$SOC=BD\times C\%\times D\times(1-F)$

Where:

- **BD**= Apparent density of soil (Mg/m^3 or g/cm^3)
- **C%**= Percentage of organic carbon in soil (%)
- **D**= Soil sampling depth (m)
- **F**= Volume fraction of coarse fragments (> 2 mm)

Parameters required for the calculation

- **BD** (soil bulk density, g/cm^3 or Mg/m^3)
- **C%** (soil organic carbon content, %)
- **D** (sample depth, m)
- **F** (percentage of coarse fragments)

4. Total Carbon Stock (CS)

Stored carbon is calculated by multiplying biomass by the biomass to carbon conversion factor (**CF** = 0,47):

$CSbiomassa=(AGB+BGB)\times CF$
$CStotale=CSbiomassa+SOC$

Where:

- **CF**= Carbon Fraction (usually 0.47 for wood and plant biomass)
- **SOC**= Soil carbon stock (see above)

Parameters required for the calculation

- **General Terms and Conditions (GTC) and the German Civil Code (BGB)** (see above)
- **CF** (Biomass to carbon conversion, default value 0.47)
- **SOC** (see above)

5. Bulk Density (BD)

The apparent density of the soil is given by:

BD=Soil dry weight\Sample volume

Where:

- **Soil dry weight** (g o Mg)
  Sample volume (cm^3 o m^3)

Parameters required for the calculation

- **Soil dry weight** (g o Mg)
  Volume of the sampling cylinder (cm^3 o m^3)

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6. Carbon Reduction and Absorption

The absorption of CO_2 by trees and soil can be calculated as:

$\Delta CO2 = 44/12 \times \Delta C$

Where:

- **44/12**= Conversion factor from C to CO_2
- **ΔC\Delta C**= Change in carbon stock between two measurements

Net CO_2 absorption is obtained as the difference between**final and initial carbon stock**:

$\Delta C = CSt1 - CSt0$

Where:

- **CSt1**= Carbon stock at time t1t1 (end of monitoring period)
- **CSt0**= Carbon stock at time t0t0 (beginning of monitoring period)

Parameters required for the calculation

- **Carbon stock at the beginning of the period (CSt0)**
- **Carbon stock at the end of the period (CSt1)**

Summary of Required Parameters

Parameter	Symbol	Unit
Diameter at chest height	DBH	cm
Tree height	H	m
Tree volume	In	m^3
Form factor	F	-
Wood density	r	g/cm^3
Biomass Expansion Factor	BEF	-
Root-to-Shoot Ratio	R	-
Percentage of organic carbon	C%	%
Soil bulk density	BD	Mg/m^3
Sample depth	D	m
Volume of the sampling cylinder	-	cm^3
Soil dry weight	-	g
Carbon stock at initial time	CS_0	Mg C/ha
Carbon stock at end time	CS_1	Mg C/ha

ANNEX 4 - MONITORING METHODOLOGIES

The **TCR Standard** adopts a comprehensive and detailed methodology for monitoring and assessing environmental impacts, specifically for ecosystem services provided by nature-based solutions, within the voluntary carbon market. This methodology integrates the **remote monitoring (remote sensing)** with data **on-site (field measurements)** to ensure accuracy and scalability.

The **remote monitoring:** It is the core of the standard's automated process, which begins with the selection and preprocessing of remotely sensed images. Various satellite data sources are used: **multispectral optical imaging** from Sentinel-2 and Landsat-8, and **immagini Synthetic Aperture Radar (SAR)** from Sentinel-1, PALSAR/PALSAR-2, with future support from NISAR. These images are carefully selected during the dry season to reduce the impact of humidity and cloud cover, and undergo advanced preprocessing, including cloud filtering for optical data and dedicated SAR pipelines for the generation of PolSAR indicators and vegetation indices. Subsequently, a **deep neural network model** trained on a large dataset (e.g., 60 million hectares of forest with aerial LiDAR data), it performs inference to estimate primary indicators such as canopy height, tree height, canopy cover, and aboveground live biomass. From these primary indicators, secondary indicators such as forest cover, belowground biomass, total biomass, carbon stocks, and their CO2 equivalents are then derived using **allometric relationships**, biomass specifications and definitions recommended by Verra and IPCC.

The **on-site monitoring (field measurements):** is crucial for the **post-calibration** of the models of the **TCR Standard**, improving the accuracy of results. User-provided field data includes detailed information on the physical and biological characteristics of trees, shrubs, or bushes within georeferenced plots. This data, collected by qualified technicians through direct measurements (e.g., relascope, clinometer, diameter tape) or well-documented estimates, includes details at the individual tree level (such as GPS location, species, tree height, diameter at breast height (DBH), canopy area, aboveground and belowground biomass) and at the sample plot level (measurement date, coordinates/geometry, main tree species, average tree height, average DBH, aboveground and belowground biomass per hectare). Model validation is performed through comparisons with global and independent reference datasets, such as GEDI L2A, GEDI L4B, Global Forest Canopy Height, and AGB data derived from airborne LiDAR and PALSAR.

The **data notarization** in the **TCR Standard (Trusted Carbon Reduction):** It is a crucial process that makes use of the **blockchain technology** to ensure the integrity and transparency of the carbon credits generated. This technology ensures the **transparency, security and immutability** information along the entire value chain, from the monitoring phase to the issuance and retirement of credits.

The application of blockchain for notarization manifests itself in several phases:

- **Approval of Methodologies**: Approval of project methodologies and issuance of credits are managed through**Smart Contracts**encoded on the blockchain. These contracts self-execute when pre-established criteria are met, automating the process and creating an **immutable registry** of approval that is transparent and resistant to manipulation.
- **Monitoring Data Integration:** Monitoring data, including potentially from IoT sensors, can be integrated with blockchain technology for a **continuous and real-time verification** of project activities. This allows for greater efficiency in auditing by third-party verifiers, who can inspect the blockchain ledger to validate the issuance and transfer of credits.
- **Issuing and Withdrawing Credits:** Once a project's carbon sequestration has been independently verified, carbon credits are issued **automatically via Smart Contracts**, with each credit representing one ton of CO2 equivalent. Subsequently, the blockchain also facilitates the **notarized withdrawal of credits**, creating a verifiable and tamper-proof record of credits withdrawn at the time of use.
- **Public Registry and Transparency**: a **blockchain-based public ledger.** It tracks all credits issued and retired, providing a transparent and easily accessible record of carbon credit activity. This ensures that all stakeholders can verify the creation, ownership, and use of credits, **significantly reducing the risk of double counting** and promoting ownership transparency.

ANNEX 5 - PROPOSTA CHECKLIST AUDIT

GHG Project Validation and Verification Checklist

I. Ex-Ante Controls (Validation)

1. **General Project Requirements and Documentation:**

   - ☐ The project design, baseline scenario, and monitoring plan are assessed to be realistic, credible, and compliant with ISO 14064-2.
   - ☐ The project provided comprehensive information, including activity type, location, geographic boundaries, start and end dates, projected aggregate GHG reductions and removals, risk identification, roles and responsibilities, environmental impacts, stakeholder consultation outcomes, and GHG program eligibility information.
   - ☐ The project meets the applicability conditions listed in the methodology.
   - ☐ All documentation needed to demonstrate project compliance with requirements is available and consistent with verification and validation needs.

2. **Baseline Scenario:**

   - ☐ The baseline is clearly stated and justified as a "business as usual" (BAU) scenario, based on historical data, industry trends, and reliable sources.
   - ☐ The baseline description includes prevalent conventional farming practices, such as continuous cropping systems, monoculture, bare-soil ploughing, use of the ploughshare, residue removal, and application of inorganic nitrogen fertilizers.
   - ☐ The start year of carbon fixation is declared.
   - ☐ The functional equivalence of the type and level of activity of the products or services offered between the project and the reference scenario is demonstrated.
   - ☐ Baseline assumptions, values, and procedures are conservatively selected to ensure that emission reductions or increases in GHG removals are not overestimated.
   - ☐ The baseline for management improvement interventions is calculated on a minimum historical basis of 5 years preceding the project.

3.

4. **Additionality:**

- ☐ It has been demonstrated that the project generates a fixation and a consequent reduction of emissions in the project area, leading to an overall improvement in the situation compared to the scenario without the project.
- ☐ The project and activities are not tied to any mandatory legal requirements for the study area and are conducted on a voluntary basis.
- ☐ The absence of intentional deforestation at the project site in the five years preceding the start of the project or the launch of additional activities is confirmed.
- ☐ The project is proven to outperform "business as usual" situations for similar projects.
- ☐ Financial additionality is demonstrated through a balance sheet that highlights a benefit derived exclusively from the sale of carbon credits.
- ☐ Where applicable, continuous measurement systems for additionalities are defined, with documentation of their compliance and periodic calibration.
- ☐ A list of reference chemicals and chemical-physical parameters to be monitored for additionality tests is defined.

5. **GHG Sink and Source (SSR):**

- ☐ All GHG SSRs controlled, related to, or influenced by the project have been identified and assessed.
- ☐ The GHG SSRs relevant to the baseline scenario are identified and compared with those of the project.
- ☐ The inclusion or exclusion of carbon pools (living biomass, waste/scraps, deadwood, soil organic carbon (SOC), harvested wood products) is justified.
- ☐ The inclusion or exclusion of reservoir-associated GHGs (CO2, CH4, N2O) is justified.
- ☐ The failure to select any GHG SSRs for regular monitoring is justified.

6. **Quantification and Methodologies:**

- ☐ Criteria, procedures or methodologies are selected or established to quantify GHG emissions and/or removals, separately for each GHG and SSR relevant to the project and the baseline scenario.
- ☐ GHG emission or removal factors are derived from a recognized source, are appropriate, current, and take into account quantification uncertainty.
- ☐ The amount of each GHG type is converted into CO2e units using the appropriate GWPs (e.g. CH4 = 28, N2O = 265 according to IPCC AR5).
- ☐ Soil carbon models are used to estimate the increase in carbon sequestration.

- ☐ The tree plantations use only vegetation types native to the project area.
- ☐ Agroforestry and orchard projects use exclusively non-invasive species (native species are preferred).
- ☐ No prior forms of carbon offset units (offsets, credits, tokens) have been created or received for the assets designated for tokenization, in order to avoid double counting.

7. **Risk Assessment for Permanence:**

- ☐ The Verra "AFOLU Non-Permanence Risk Tool" Version 4.2 methodology was applied to assess internal, external and natural risk.
- ☐ The overall risk score is less than 60, and the scores for the individual risk categories (internal <35, external <20, natural <35) are respected, confirming the robustness of the project (e.g. total score 39.50).
- ☐ The project includes a minimum 10% credit buffer to ensure permanence, or a non-permanence risk mitigation strategy is documented and implemented if the buffer is lower.
- ☐ Ownership of the credits is defined in advance and unambiguously for the 5-year certification period.
- ☐ A written consent from the owner of the area is required if the owner of the credits is not the same as the owner of the management area.
- ☐ Proactive measures are envisaged to safeguard the integrity of carbon offsetting activities, such as planting an additional 10% of each species or allocating 10% without tokenization.

8. **Initial Monitoring Plan:**

- ☐ A monitoring plan is established that includes objectives, measured parameters, data types, sources, methodologies (estimation, modeling, measurement, calculation, uncertainty), frequency, roles, responsibilities, controls, and GHG information management systems, including equipment calibration.
- ☐ Data quality management procedures suitable for information management are defined, including uncertainty assessment, to reduce uncertainty levels.
- ☐ The roles and responsibilities of the personnel involved in the project activities and in the verification/control processes are specified.
- ☐ The sampling and measurement methodologies are statistically appropriate and in accordance with the IPCC Guidelines (e.g. minimum depth of sampling of soil/seabed mud of 30 cm).
- ☐ The sampling design aims to locate sample points in each selected layer (e.g. using TWI, NDVI, soil type for K-means stratification).
- ☐ The sample size is determined to achieve a required precision of at least 15% at a 95% confidence level.

- ☐ The procedures for soil sampling and analysis for soil organic carbon (SOC) according to the Walkley & Black standard method, to be carried out by an accredited laboratory, are defined.
- ☐ Accurate historical land use and land cover (LULC) data are available for at least three points in time (0-3, 4-9, 10-15 years before project initiation).

II. Ex-Post Controls (Verification)

1. **Effective Performance Evaluation:**

- ☐ Actual GHG emission reductions and carbon sequestration are assessed against the project baseline and monitoring reports.
- ☐ The project's performance and ongoing impact are evaluated annually.
- ☐ Emission reductions for the year (ERy) are calculated as the difference between Baseline Emissions (BEy) and Project Emissions (PEy).
- ☐ The net benefit of carbon removal is quantified as CRbaseline – CRtotal – GHGincrease.
- ☐ Total changes in carbon stocks (CRtotal) are calculated based on measurements of carbon sinks (SOC, living biomass, harvested wood products) at two points in time.

2. **Monitoring and Data Collection:**

- ☐ GHG monitoring procedures are applied in accordance with the established monitoring plan.
- ☐ Continuous internal monitoring is performed annually, ensuring the implementation of the proposed practices.
- ☐ Monitoring is carried out by the Certification Body (CB) on an area equal to at least 5% of the total certified area.
- ☐ All data and information related to project monitoring are recorded and documented.
- ☐ Carbon sequestration is monitored through soil sampling and analysis to measure changes in SOC content, conducted at regular intervals (every 3-5 years or prior to each audit) on a representative sample of the project area.
- ☐ A sophisticated mix of soil sampling and advanced remote sensing technologies (e.g. Sentinel-1, Sentinel-2, Landsat 8/9 satellite imagery) is used.
- ☐ Remote sensing is used to monitor vegetative health, land cover changes, and biomass accumulation.

- ☐ Satellite imagery is used to monitor the implementation of regenerative agricultural practices, such as crop presence/removal, cover crop use, and tillage.
- ☐ Measurements of diameter at breast height (BDH) and height of sampled trees are performed to estimate woody biomass.
- ☐ Litter biomass is measured and its dry weight is extrapolated per hectare.
- ☐ GHG emissions are collected using calculation methodologies based on the type and quantity of fertilizers and pesticides used, and measurements of soil carbon sequestration.
- ☐ Data are collected through a multi-level approach, combining direct data from farms (questionnaires, interviews, registers) with data from secondary sources (databases, literature).
- ☐ The measuring and monitoring equipment has been properly calibrated, tested and maintained.
- ☐ Soil samples are taken in accordance with internal protocol and analyzed by an accredited laboratory, with copies of laboratory reports and evidence of sample location retained.

3. **Reporting and Transparency:**

- ☐ Detailed annual reports on project progress, including GHG emission reductions, carbon sequestration, and the adoption of sustainable practices, are compiled and submitted.
- ☐ Each monitoring report is prepared and published in the public register.
- ☐ Any deviations from the certification project (e.g., losses due to adverse weather conditions) are reported by updating the public register.
- ☐ The monitoring report includes changes in CO2 fixation, changes in CO2 emissions, leakage extent reporting, and recalculation of the residence buffer.
- ☐ In case of using automatic detection stations, specific summary tables of the temporal analyses are presented.
- ☐ The summarized data and results are made available on the project website to ensure transparency and provide insights into the project's impact.
- ☐ A GHG report is prepared and made available to intended users, identifying the use/user, format/content, and including a statement of third-party verification or validation.

4. **Quality Management and Continuous Risk:**

- ☐ Quality Assurance/Quality Control (QA/QC) procedures are implemented, including staff training, equipment calibration, and independent review and verification of monitoring data.
- ☐ The uncertainty associated with estimating emissions and removals from the agricultural system is continuously assessed.

- ☐ The permanence buffer is recalculated if remeasurements of soil organic carbon stock indicate lower stocks than previously estimated by the model.
- ☐ Ownership of the credits is modified if the lessee changes during the five-year certification period or at the end of the first five-year period.

Project Design Document (PDD) Validation Template - TCR Methodology

This document serves as a reference for validating a Project Design Document (PDD) that intends to adhere to the Trusted Carbon Reduction (TCR) methodology. Validation aims to confirm that the PDD complies with the requirements established by the TCR methodology and relevant international standards, including ISO 14064-2, the IPCC Guidelines, Verra VM0042, and CDM AR-AMS0007.

Validation Date:[Insert Date]

Validator Name:[Validator Name]

PDD Reference:[PDD ID/Name]

PDD Version:[PDD Version]

Validation Phase	Request	Answer (Yes/No)	Notes
1. General Information and Project Suitability			
1.1	Does the PDD provide a comprehensive description of the project, including type of activities, goals and objectives, in line with the TCR methodology?		
1.2	Have the essential details of the applicant and the project representative been provided, along with relevant identity proof (KYC/KYB)?		
1.3	Does the project demonstrate long-term legal title or right to the land for a minimum of 20 years from the onboarding date, with verifiable documentation?		
1.4	Has documented proof of purchase of the trees or plants been provided, where the business requires it?		
1.5	Have the required images and/or videos been included for each plant/tree species involved in the additional activities?		
1.6	Does the project explicitly commit to maintaining green assets and carbon offsets for a minimum of 20 years from the onboarding date?		
1.7	Does the PDD provide convincing evidence that the CO_2 offsetting activities would not have occurred in the absence of the project intervention (Additionality)?		

1.8	Has the project site not undergone deliberate deforestation in the five years preceding the project start date or the commencement of additional activities?		
1.9	For the planting projects, were only native vegetation and/or non-invasive species used, where specified?		
1.10	Does the project declare that it has not previously created or received carbon offsets (e.g., offsets, credits, tokens) for the same activities, thus avoiding double counting?		
2. System Definition and SSR (Sources, Absorbers and Reservoirs)			
2.1	Does the PDD clearly define the system (farm or group of farms) and time frame for analyzing farming activities?		
2.2	Have all relevant SSRs (controlled, correlated, unaffected) been identified and classified, including those influenced by agricultural activities (e.g. machinery emissions, N2O from fertilizers, soil/biomass sequestration)?		
2.3	Are the included carbon sinks (Living Biomass, Dead Wood, Soil Organic Carbon, Harvested Wood Products) consistent with the TCR methodology specifications?		
2.4	Are the included (CO2 from SOC, N2O from fertilizers/nitrogen-fixing species) and excluded (CO2 from fossil fuels, CH4 from methanogenesis/enteric fermentation/manure, biomass combustion) GHG sources justified and compliant with the methodology?		
3. Determination of Baseline and Additionality			
3.1	Is the baseline defined as the “business as usual” (BAU) scenario based on conventional farming practices and relevant historical data?		
3.2	Are the baseline quantification methodologies (IPCC Guidelines, Verra VM0042) applied correctly, taking into account emission factors and soil characteristics?		
3.3	Has an assessment of the uncertainty associated with the baseline estimate been performed?		

3.4	Does the PDD demonstrate that mitigation interventions generate net GHG reductions/removals that are additional to the baseline (positive “Additionality” value)?		
3.5	Does the PDD justify the lack of pre-existing incentives that would have led to the adoption of sustainable practices in the absence of the project?		
4. Quantifying GHG Reductions/Removals			
4.1	Do data collection procedures combine direct (farm data) and secondary (database, literature) sources to ensure accuracy?		
4.2	Are the quantification standards (LIFE C-Farms, Verra VM0042, CDM AR-AMS0007, IPCC Guidelines) correctly applied to estimate the project emissions/removals?		
4.3	Has an assessment been performed of the uncertainty associated with the estimation of emissions/removals from the agricultural system?		
4.4	The calculation of the Net Carbon Removal Benefit follows the formula: CRbaseline – CRtotal – GHGincrease?		
4.5	Do the calculations of changes in carbon stocks in the pools (SOC, Living Biomass, Harvested Wood Products) follow the indicated formulas and requirements?		
4.6	Is the GHG increase calculated correctly, comparing project emissions to baseline emissions and including only differences > 0?		
4.7	Does the PDD consider and evaluate “leakage” (movement of emissions outside the project boundaries)?		
4.8	Has a buffer of 10% of the carbon removal units been set aside to cover any future losses?		
4.9	Are the quantification estimates for specific agricultural practices (e.g., Organic Farming, Conservation Tillage, Cover Crops, etc.) consistent with the values and sources indicated in the TCR methodology?		
5. Monitoring and Reporting			

5.1	Is the monitoring plan clearly defined, including frequency, methodologies, indicators, and responsibilities?		
5.2	Does the PDD describe the use of data collection methods (in-situ, remote sensing, database) and related procedures to ensure reliability?		
5.3	Are the procedures for analyzing monitoring data (trend assessment, anomaly identification) appropriate?		
5.4	Are reporting procedures (frequency, clarity, compliance with standards) compliant with TCR requirements?		
5.5	Does the PDD require independent third-party verification to ensure the accuracy and credibility of the data?		
6. Risk Assessment for Permanence			
6.1	Was the non-permanence risk assessment conducted using the Verra "AFOLU Non-Permanence Risk Tool" Version 4.2?		
6.2	Have internal, external and natural risks been assessed, and does the overall score and the score for each category meet the established limits (overall ≤ 60, internal ≤ 35, external ≤ 20, natural ≤ 35)?		
6.3	Has the project implemented proactive measures to mitigate tree damage (e.g., 10% additional planting or non-tokenized allocation)?		
7. Compliance with Standards and Adopted Methodologies			
7.1	Does the PDD demonstrate compliance and integration with the principles and requirements of ISO 14064-2 for quantification, monitoring and reporting at the project level?		
7.2	Is the use of the IPCC Guidelines for National Greenhouse Gas Inventories documented and applied correctly for estimating emissions/removals?		
7.3	Is the Verra VM0042 methodology integrated for the quantification of GHG removals associated with soil carbon sequestration?		
7.4	Are the principles and procedures of the CDM AR-AMS0007 methodology integrated for the quantification of emissions and		

	removals from afforestation/reforestation activities?		
7.5	Does the PDD describe how blockchain technology is used to ensure transparency, traceability, and immutability throughout the lifecycle of carbon credits?		

Final Validator Analysis:

- **General Compliance:**[Overall assessment of PDD compliance with TCR standards.]
- **Areas of Strength:**[Strengths of PDD, e.g., excellent documentation, innovative approach.]
- **Areas Requiring Improvement/Clarification:**[Weaknesses or ambiguities that require review before final validation.]
- **Recommendations:**[Tips for the project proposer.]

Validator's Signature:

[Validator Name]

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2. UNI EN ISO 14064-2:2019 Greenhouse gases - Part 2: Specification with project-level guidelines for quantifying, monitoring and reporting greenhouse gas emission reductions or removal increases.
3. UNI EN ISO 14065:2013 Greenhouse gases - Requirements for greenhouse gas validation and verification bodies for accreditation or other forms of recognition.
4. UNFCCC (2011) Methodological Tool. Tool for Demonstration and Assessment of Additionality (Version 06.0.0) CDM Executive Board [http://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v5.2.pdf/historyview]
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    - Standard VCS v3.2 (Voluntary Carbon Standard)
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